Adding missing data to Pend Oreille

The goal is to find out why there is a gap between the Pend Oreille River and the Lake on our plots, and then possibly fix it.

TL;DR

This is a working example of how to add to a GeoDataFrame. It also shows how to find nearest points and compute lengths.

In […

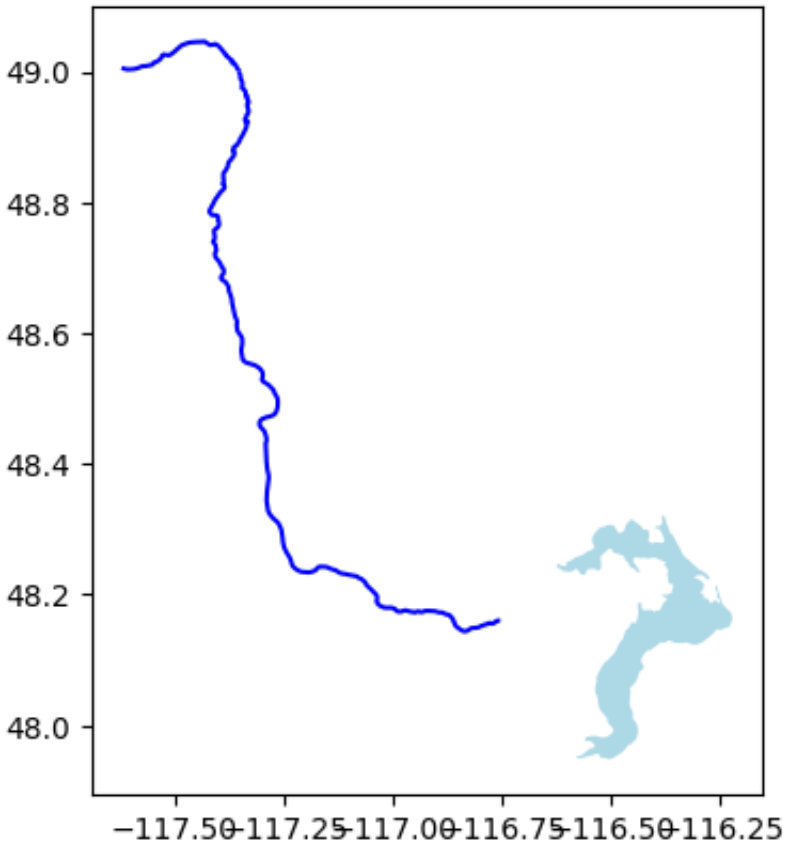
```
import geopandas as gpd
import pandas as pd
import matplotlib.pyplot as plt

dbpath = '/Users/telliott/data/'
fn = 'North_America_Lakes_and_Rivers.zip'
na_rivers = gpd.read_file(dbpath + fn)
sel = na_rivers['NameEn'].str.contains("Pend Oreil
po_river = na_rivers[sel]
```

In […

```
fn = 'North_America_Lakes.zip'
na_lakes = gpd.read_file(dbpath+fn)
sel = na_lakes['NameEn'].str.contains('Pend Oreille
po_lake = na_lakes[sel]
```

In […

```
ax = po_river.plot(color='b')
po_lake.plot(ax=ax,color='lightblue')
plt.show()
```



`po_river` has two rows, one for the part in Canada and the second for the part in the USA. `po_lake` has only a single row, a `LINESTRING` object.

In […

```
river_geom = po_river.iloc[1].geometry
lake_geom = po_lake.geometry
L = list(river_geom.coords)
len(L)
print(L[0],L[-1])
```

```
(-116.754568877487, 48.1599716677888) (-117.353364035974, 49.0003164329327)
```

The river data appears to run generally east to west. However, we should not assume that there aren't some other points in `L` that are closer. One idea is to look at the `bounds`.

In […

```
river_geom.bounds
```

Out[…
```
(-117.421307883071, 48.1439997098789, -116.754568877487, 49.0003164329327)
```

nearest_points

In […
```
from shapely.ops import nearest_points
points = nearest_points(river_geom,lake_geom)
print(points)
```

```
(6229    POINT (-116.75457 48.15997)
Name: geometry, dtype: geometry, 6229    POINT (-116.62599 48.24265)
Name: geometry, dtype: geometry)
```

I'm struggling to find the right way to get these plotted. This doesn't work

In […
```
def display(gdf):
    ax = po_river.plot(color='b')
    po_lake.plot(ax=ax,color='lightblue')
    gdf.plot(ax=ax)
    plt.show()

# display(points)
```

In […
```
print(type(points))
```

```
<class 'tuple'>
```

In […
```
points[0]
```

Out[…
```
6229    POINT (-116.75457 48.15997)
Name: geometry, dtype: geometry
```

In […
```
points[1]
```

Out[…
```
6229    POINT (-116.62599 48.24265)
Name: geometry, dtype: geometry
```

For the moment, do it by hand.

In […

```
D = { 'name':['river','lake'],
      'LON':[-116.75457,-116.62286],
      'LAT':[48.15997,48.24656] }
df = pd.DataFrame(D)
df
```

Out[…

	name	LON	LAT
0	river	-116.75457	48.15997
1	lake	-116.62286	48.24656

In […

```
from shapely import Point
geometry = [Point(xy) for xy in zip(df.LON, df.LAT)
gdf = gpd.GeoDataFrame(df,geometry=geometry,crs=po_
print(gdf)
```

```
    name        LON       LAT                       g
eometry
0  river -116.75457  48.15997  POINT (-116.75457 4
8.15997)
1   lake -116.62286  48.24656  POINT (-116.62286 4
8.24656)
```

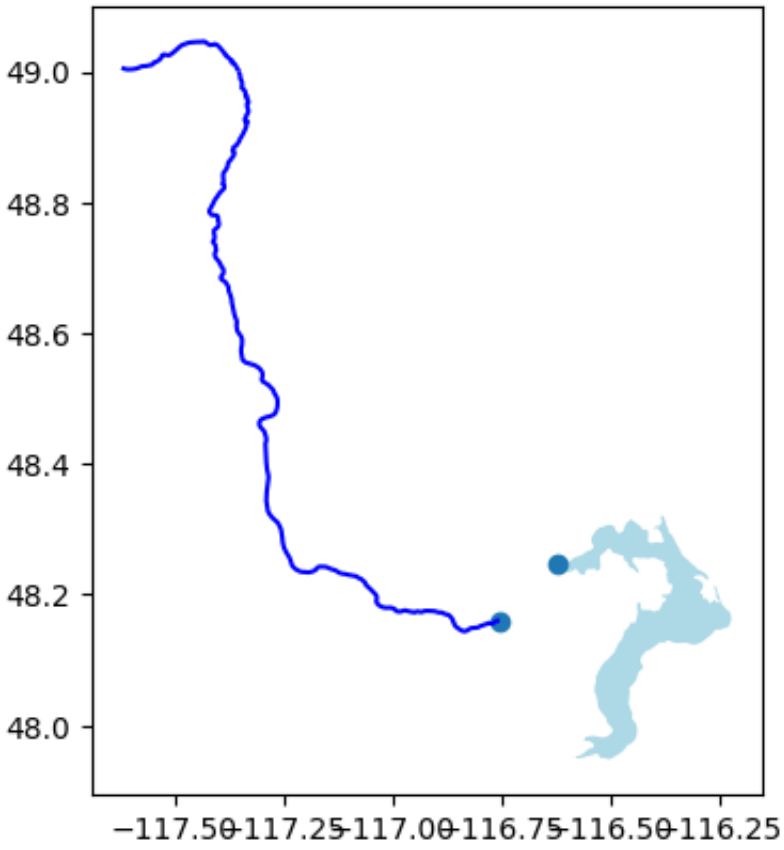
In […

```
print(gdf.columns)
```

```
Index(['name', 'LON', 'LAT', 'geometry'], dtype='ob
ject')
```

In […

```
# from above
display(gdf)
```



Size of the gap

We're missing 0.129 degrees of longitude and 0.08659 of latitude.

In […

```
# fundamental, approximate approach
from math import cos, radians

C = 40000 # km
dx = C/360 * 0.129 * cos(radians(48))
dy = C/360 * 0.08659
length = (dx**2 + dy**2)**0.5
round(length,2)
```

Out[…

```
13.58
```

Here's a better way:

In […

```
import utm
from pyproj import CRS
```

```
dbpath = '/Users/telliott/data/'
po_addns = gpd.read_file(
    dbpath + 'po_addns.shp.zip')

# to get length properly we must project to correct

def utm_crs_from_latlon(lat, lon):
    crs_params = dict(
        proj = 'utm',
        zone = utm.latlon_to_zone_number(lat, lon),
        south = lat < 0
        )
    return CRS.from_dict(crs_params)

utm_crs = utm_crs_from_latlon(48.2,-116.6)
print(utm_crs)

sub = po_addns.to_crs(utm_crs)
print(sub.geometry[0])
print(sub.length)
```

```
+proj=utm +zone=11 +type=crs
LINESTRING (526404.0644849889 5343991.683102484, 52
4411.129314948 5344830.198400886, 523166.4076062218
5344103.82374203, 522793.4136110913 5342407.191967
9, 522490.65436216904 5341197.455119945, 521066.933
7879284 5339658.441673474, 521182.88584540464 53376
73.630000758, 521185.69705673005 5336915.458459129,
520175.2114436488 5335918.793824149, 519526.6389227
794 5334923.363112617, 519075.55416763667 5334398.0
95703166, 518219.3099086929 5334112.829505766)
0    15632.707337
dtype: float64
```

That's 15632.707337 meters. Considering the curve, that seems pretty reasonable.

In […

```
### Adding the points
```

I just take points from Google Maps. Notice that these are in `(y,x)` order, because that's how they roll.

In […
```
raw_points = [
[48.248589733216136, -116.64432444663814],
[48.256213458503204, -116.67112139993714],
[48.24972525370046, -116.68793039791562],
[48.23447473170333, -116.69304617990908],
[48.22360190290814, -116.69718752629615],
[48.209804691843075, -116.71643261093814],
[48.19194395801155, -116.71497095894],
[48.185122762281175, -116.71497095894],
[48.17618887113204, -116.7286130442559],
[48.167253423001256, -116.73738295624466],
[48.162541377351445, -116.74347317265425],
[48.159235, -116.749974]]
len(raw_points)
```

Out[… 12

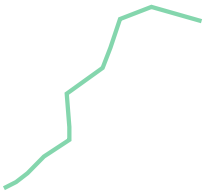
I am sure there is a better way to do this part. I just copy from the original.

In […
```
points = [(b,a) for a,b in raw_points]
```

In […
```
D = {'FID':['4995'],
'Country':['USA'],
'NameEn':['Pend Oreille River'],
'NameEs':['Río Pend Oreille'],
'NameFr':['Pend Oreille River'],
'LengthKm':['15.632707'] }

df = pd.DataFrame(D)
```

In […
```
from shapely import LineString
# the geometry part
LS = LineString(points)
LS
```

Out[…



In […

```
new_pts = gpd.GeoDataFrame(
    # notice [LS]
    df,geometry=[LS],crs="EPSG:4326")

new_pts.to_file('po_addns.shp.zip',
    driver='ESRI Shapefile')
```

In […

```
new_pts.crs
```

Out[…

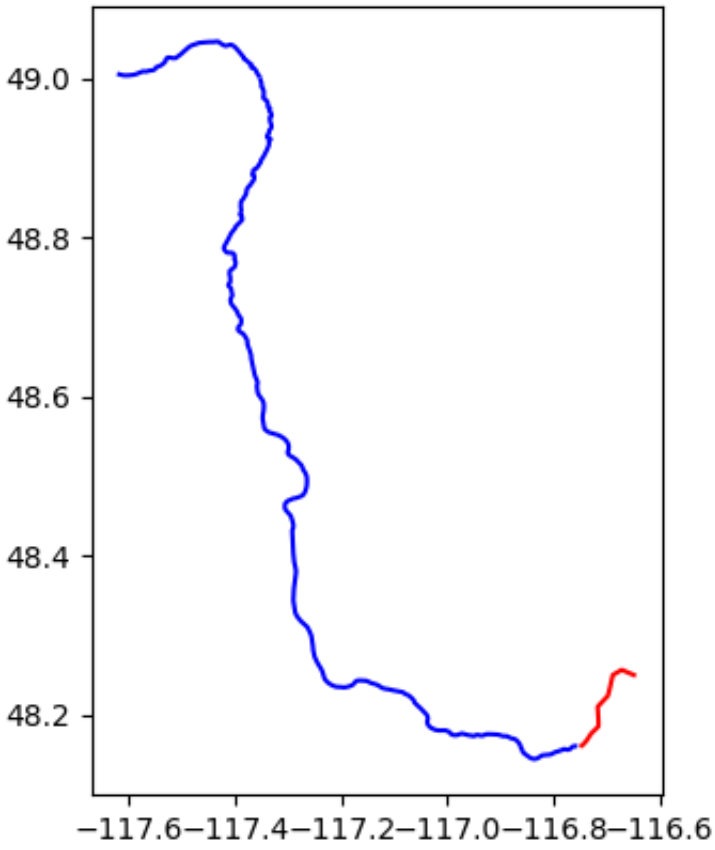
```
<Geographic 2D CRS: EPSG:4326>
Name: WGS 84
Axis Info [ellipsoidal]:
- Lat[north]: Geodetic latitude (degree)
- Lon[east]: Geodetic longitude (degree)
Area of Use:
- name: World.
- bounds: (-180.0, -90.0, 180.0, 90.0)
Datum: World Geodetic System 1984 ensemble
- Ellipsoid: WGS 84
- Prime Meridian: Greenwich
```

The script to write and plot the new GeoDataFrame is here.

And here is the plot:

In […

```
ax = po_river.plot(color='b')
new_pts.plot(ax=ax,color='r')
plt.show()
```



In […